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|------------------|--------------------------------------------|
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| <b>Duration</b>  | 58 mins 18 secs                            |
| <b>Grade</b>     | <b>17.50</b> out of 20.00 ( <b>87.5%</b> ) |

### Question 1

Partially correct

Mark 0.67 out of 1.00

Which statements about attention mechanism are true?

- ☒ a. An LSTM with self-attention has access to all previous hidden states and cell states via an softmax-based attention mechanism. ✓
- ☐ b. The LSTM gates can be interpreted as an attention mechanism over the input features.
- ☒ c. If we apply the softmax function to the attention score  $e(i)$  we obtain the attention vector  $a(i)$ . ✓
- ☐ d. A spatial attention mechanism assigns equal weight to each pixel in an image.
- ☐ e. The scaled dot-product attention mechanism generally has the form  $\text{softmax}\left(\frac{1}{\sqrt{d}}VK^\top\right)Q$ , where  $V, K, Q \in \mathbb{R}^{L \times d}$  are values, keys, and queries, respectively, and  $L$  is the sequence length and  $d$  is the hidden dimension.

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: The LSTM gates can be interpreted as an attention mechanism over the input features., An LSTM with self-attention has access to all previous hidden states and cell states via an softmax-based attention mechanism., If we apply the softmax function to the attention score  $e(i)$  we obtain the attention vector  $a(i)$ .

### Question 2

Correct

Mark 1.00 out of 1.00

Which statements about LSTM applications are true?

- ☐ a. Radford et al. (2017) used an LSTM to discover sentiment in Amazon reviews by training it for sequence classification (good vs. bad) on the review text.
- ☒ b. A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let  $f(\cdot)$  be the (trained) embedding function, then  $f(\text{queen}) - f(\text{woman}) + f(\text{man})$  is found close to the point  $f(\text{king})$ . ✓
- ☐ c. Fan et al. (2015) trained an LSTM to predict audio from videos of speeches. Given a muted video recording of a speech, they could accurately recover the sound of the speech.
- ☐ d. LSTM can be used for protein classification. Hereby, the protein is represented as a sequence of atoms and the labels are either functional or structural properties of the respective protein.
- ☒ e. "word2vec" (Mikolov et al., 2013) is an example for a word embedding technique. ✓

Your answer is correct.

The correct answers are: A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let  $f(\cdot)$  be the (trained) embedding function, then  $f(\text{queen}) - f(\text{woman}) + f(\text{man})$  is found close to the point  $f(\text{king})$ ., "word2vec" (Mikolov et al., 2013) is an example for a word embedding technique.

**Question 3**

Partially correct

Mark 0.67 out of 1.00

The Elman Network (without output activation function and bias units) can be defined as

$$s(t) = Wx(t) + a(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors  $x(t)$ , hidden pre-activation vectors  $s(t)$ , hidden activation vectors  $a(t)$ , activation function  $f(\cdot)$ , and parameter matrices  $W, V$ . Which of the following statements are true?

- ☐ a. The Elman network is well suited for training with RTRL because the recurrent Jacobian has a diagonal structure and this reduces the computational complexity.
- ☒ b. The recurrent weights are equal to the identity matrix. ✓
- ☐ c. The hidden units of the Elman network are interconnected, i.e., at time  $t$  each hidden unit receives input from all other hidden units at time  $t-1$ .
- ☒ d. The recurrent Jacobian of the hidden activations is  $\frac{\partial a(t)}{\partial a(t-1)} = \text{diag}(f'(s(t)))$ . ✓
- ☐ e. The matrix  $W$  contains the recurrent weights.

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: The recurrent weights are equal to the identity matrix., The recurrent Jacobian of the hidden activations is  $\frac{\partial a(t)}{\partial a(t-1)} = \text{diag}(f'(s(t)))$ ., The Elman network is well suited for training with RTRL because the recurrent Jacobian has a diagonal structure and this reduces the computational complexity.

**Question 4**

Correct

Mark 1.00 out of 1.00

Which of the following statements about RNNs are true?

- ☒ a. The number of RNN parameters is independent of the sequence length. ✓
- ☐ b. One advantage of RNNs is that their computations can be easily parallelized over multiple time steps.
- ☐ c. A disadvantage of RNNs is that they can only be applied to sequences of a fixed length.
- ☐ d. Many RNN architectures are Turing complete. This means that they can be trained to exactly imitate the behavior of any computer program.
- ☒ e. RNNs update an internal state variable at each time step. ✓

Your answer is correct.

The correct answers are: RNNs update an internal state variable at each time step., The number of RNN parameters is independent of the sequence length.

**Question 5**

Partially correct

Mark 0.75 out of 1.50

The LSTM (without bias units and forget gate) is defined as

$$z(t) = \tanh(Ux(t) + Ph(t-1))$$

$$i(t) = \sigma(Vx(t) + Qh(t-1))$$

$$c(t) = c(t-1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t-1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors  $x(t)$ , hidden activation vectors  $h(t)$ , memory cell state vectors  $c(t)$ , gate activation vectors  $z(t), i(t), o(t)$ , weight matrices  $P, Q, R, U, V, W$ . Let  $L(t) = L(y(t), \hat{y}(t))$  denote the loss at time  $t$  and let  $L = \sum_{t=1}^T L(t)$  denote the total loss. We use denominator-layout convention, i.e.,  $\frac{\partial L}{\partial c(t)}$  is a column vector. The diag operator turns a vector into a diagonal matrix, i.e.,  $\text{diag}((1, 1)^\top) = I \in \mathbb{R}^{2 \times 2}$  and  $\odot$  denotes Hadamard's product. Which of the following statements are true?

- ☐ a. The gradient of the loss with respect to the hiddens is  $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)} + U \frac{\partial L}{\partial z(t-1)} + V \frac{\partial L}{\partial i(t-1)} + R \frac{\partial L}{\partial o(t-1)}$ .
- ☐ b. Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.
- ☐ c. The LSTM architecture has no exploding or vanishing gradients because  $\frac{\partial h(t)}{\partial h(t-1)}$  is a diagonal matrix.
- ☐ d. If we adapt  $z(t) = \sigma(Ux(t) + Ph(t-1))$ , then  $z(t) \geq 0$  for all  $t$  and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.
- ☒ e. The memory cell fulfills  $\frac{\partial c(t)}{\partial c(t-\tau)} = I$  (neglecting dependencies via the hiddens) for any  $\tau \in \{1, \dots, t-1\}$ , where  $I$  is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel. ✓

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: The memory cell fulfills  $\frac{\partial c(t)}{\partial c(t-\tau)} = I$  (neglecting dependencies via the hiddens) for any  $\tau \in \{1, \dots, t-1\}$ , where  $I$  is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel., If we adapt  $z(t) = \sigma(Ux(t) + Ph(t-1))$ , then  $z(t) \geq 0$  for all  $t$  and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.

**Question 6**

Correct

Mark 1.00 out of 1.00

Which statements about the gated recurrent unit (GRU, Cho et al., 2014) are true?

- ☐ a. The so-called "update gate" of the GRU acts as a coupled input and output gate.
- ☒ b. The GRU has one gate less than the fully featured LSTM architecture. ✓
- ☐ c. The GRU does not have any form of forget-gate mechanism.
- ☒ d. The standard GRU has less parameters than a fully featured LSTM (i.e., with forget gate) that has the same number of hidden units. ✓
- ☒ e. In the GRU, a so-called "reset gate"  $r(t)$  regulates how strongly the increments  $z(t)$  depend on the old hidden  $h(t - 1)$ . ✓

Your answer is correct.

The correct answers are: The GRU has one gate less than the fully featured LSTM architecture., The standard GRU has less parameters than a fully featured LSTM (i.e., with forget gate) that has the same number of hidden units., In the GRU, a so-called "reset gate"  $r(t)$  regulates how strongly the increments  $z(t)$  depend on the old hidden  $h(t - 1)$ .

**Question 7**

Correct

Mark 1.50 out of 1.50

Backpropagation through time (BPTT) is the standard gradient computation algorithm for RNNs. However, there exist alternatives, e.g., truncated BPTT and real-time recurrent learning (RTRL). Which statements about these algorithms are true.

- ☒ a. RTRL computes the same gradients as BPTT ✓
- ☒ b. Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them. ✓
- ☒ c. RTRL is very slow and therefore rarely used in practice. ✓
- ☐ d. Truncated BPTT has advantages if the input sequences are short.
- ☐ e. The runtime complexity of RTRL is  $O(N^3)$  per time step, where  $N$  denotes the number of hidden units.
- ☐ f. Truncated BPTT computes the same gradients as BPTT.

Your answer is correct.

The correct answers are: Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them., RTRL computes the same gradients as BPTT, RTRL is very slow and therefore rarely used in practice.

**Question 8**

Partially correct

Mark 0.67 out of 1.00

Which statements about LSTM variants are true?

- ☒ a. Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos. ✓
- ☐ b. Stacked LSTM (Graves et al., 2013) stack LSTM layer by using one LSTM's final cell state as the initial cell state for the next LSTM.
- ☐ c. A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction.
- ☒ d. Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state. ✓
- ☐ e. The key idea of multidimensional LSTM (Graves et al., 2007) is to enable the network to process sequences consisting of vectors which stem from high-dimensional vector spaces.

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state., A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction., Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos.

**Question 9**

Correct

Mark 1.00 out of 1.00

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors  $x(t)$ , hidden pre-activation vectors  $s(t)$ , hidden activation vectors  $a(t)$ , activation function  $f(\cdot)$  and parameter matrices  $R, W, V$ . Let  $\theta$  denote the vector of all network parameters shared in time and let  $\theta(t)$  denote their usage at time  $t$ . Further, let  $L(t) = L(y(t), \hat{y}(t))$  denote the loss function at time  $t$  and let  $L = \sum_{t=1}^T L(t)$  denote the total loss. We use denominator-layout convention, i.e.,  $\frac{\partial L}{\partial \theta}$  is a column vector.

Which statements about RTRL are true?

- ☒ a. BPTT considers the recursion  $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L(t)}{\partial s(t)}$  while RTRL considers the recursion  $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t-1)}{\partial \theta} \frac{\partial s(t)}{\partial s(t-1)}$ . ✓
- ☒ b. RTRL computes the gradients  $\frac{\partial L(t)}{\partial \theta}$  during the forward pass already. ✓
- ☐ c. The term  $\frac{\partial s(t)}{\partial \theta}$  generally has  $O(N^4)$  elements, where  $N$  denotes the number of hidden units
- ☐ d. Schmidhuber's approach divides the input sequence into chunks of size  $N$  (number hidden units) and performs RTRL within these chunks. Then it uses BPTT to consolidate the gradients for these chunks.

Your answer is correct.

The correct answers are: BPTT considers the recursion  $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L(t)}{\partial s(t)}$  while RTRL considers the recursion  $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t-1)}{\partial \theta} \frac{\partial s(t)}{\partial s(t-1)}$ , RTRL computes the gradients  $\frac{\partial L(t)}{\partial \theta}$  during the forward pass already.

**Question 10**

Correct

Mark 1.50 out of 1.50

Which statements about the xLSTM are true?

- ☒ a. Exponential gating addresses the limitation of LSTMs regarding their inability to revise storage decisions - it allows xLSTM to scale down older inputs (revise past storage decisions) ✓
- ☒ b. The mLSTM cell doesn't have recurrent connections. ✓
- ☐ c. The memory capacity of xLSTM increases with the sequence length.
- ☒ d. The mLSTM variant of xLSTM uses a matrix memory and covariance update rule. ✓
- ☐ e. The sLSTM cell doesn't have recurrent connections.
- ☒ f. The sLSTM memory is independent of sequence length and hidden dimension. ✓
- ☒ g. The mLSTM matrix memory size is independent of sequence length. ✓

Die Antwort ist richtig.

The correct answers are: Exponential gating addresses the limitation of LSTMs regarding their inability to revise storage decisions - it allows xLSTM to scale down older inputs (revise past storage decisions), The mLSTM variant of xLSTM uses a matrix memory and covariance update rule., The sLSTM memory is independent of sequence length and hidden dimension., The mLSTM matrix memory size is independent of sequence length., The mLSTM cell doesn't have recurrent connections.

**Question 11**

Correct

Mark 1.00 out of 1.00

Which statements about the self-attention mechanism are correct?

- ☒ a. In contrast to recurrent neural networks, self-attention processes input sequences in parallel. ✓
- ☐ b. The vector at position  $t$  in the output sequence of a self-attention layer is computed by summing the key vectors, each weighted by its corresponding attention score from the  $t$ -th row of the attention matrix.
- ☐ c. In the standard self-attention formulation, the euclidean distance is used as a measure of alignment between keys and queries.
- ☐ d. The Multi-head Attention mechanism consists of a total of three weight matrices.
- ☒ e. The raw attention scores are normalized by applying a softmax activation function row-wise to the attention matrix. ✓

Your answer is correct.

The correct answers are:

In contrast to recurrent neural networks, self-attention processes input sequences in parallel., The raw attention scores are normalized by applying a softmax activation function row-wise to the attention matrix.

**Question 12**

Correct

Mark 1.00 out of 1.00

Which statements about Transformer networks (Vaswani et al., 2017) are true?

- ☒ a. The algorithmic time complexity of the Transformer is quadratic in the length of the input sequence. ✓
- ☐ b. Transformer networks use multi-head attention, which works as follows. It runs multiple instances of scaled dot-product attention, builds a sum over their outputs (or context) and feeds the result into a linear layer.
- ☒ c. BERT (Devlin et al., 2019) uses two different pre-training objectives, (1) inserting missing (masked out) words into a sentence, and (2) next sentence prediction, i.e., deciding whether sentence B is the successor of sentence A. ✓
- ☐ d. Without positional encoding, the Transformer output changes if the input sequence is permuted.
- ☐ e. The Transformer architecture consists of an encoder and a decoder part, where the encoder depends on the output of the decoder.

Your answer is correct.

The correct answers are: BERT (Devlin et al., 2019) uses two different pre-training objectives, (1) inserting missing (masked out) words into a sentence, and (2) next sentence prediction, i.e., deciding whether sentence B is the successor of sentence A., The algorithmic time complexity of the Transformer is quadratic in the length of the input sequence.

**Question 13**

Correct

Mark 2.00 out of 2.00

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors  $x(t)$ , hidden pre-activation vectors  $s(t)$ , hidden activation vectors  $a(t)$ , activation function  $f(\cdot)$  and parameter matrices  $R, W, V$ . Let  $L(t) = L(y(t), \hat{y}(t))$  denote the loss function at time  $t$  and let  $L = \sum_{t=1}^T L(t)$  denote the total loss. We use denominator-layout convention, i.e.,  $\delta(t) = \frac{\partial L}{\partial s(t)}$  is a column vector. Which of the following statements are true?

- ☒ a. BPTT has to store all hidden activations  $a(1), a(2), \dots, a(T)$ , i.e., its memory demand grows with the sequence length. ✓
- ☒ b. BPTT is the standard algorithm for computing the gradients of a recurrent neural network. ✓
- ☐ c. The gradient of the loss with respect to the input weights  $W$  can be written as  $\frac{\partial L}{\partial W} = \sum_{t=1}^T x(t)\delta^\top(t)$ .
- ☐ d. The deltas fulfill the recursive relation  $\delta(t) = \text{diag}(f'(s(t))) \left( V \frac{\partial L(t)}{\partial \hat{y}(t)} + R\delta(t+1) \right)$ .
- ☐ e. The gradient of the loss with respect to the recurrent weights  $R$  can be written as  $\frac{\partial L}{\partial R} = \sum_{t=1}^T \delta(t)s^\top(t-1)$ .

Your answer is correct.

The correct answers are: BPTT is the standard algorithm for computing the gradients of a recurrent neural network., BPTT has to store all hidden activations  $a(1), a(2), \dots, a(T)$ , i.e., its memory demand grows with the sequence length.

**Question 14**

Partially correct

Mark 0.75 out of 1.50

The LSTM (without bias units and forget gate) is defined as

$$z(t) = \tanh(Ux(t) + Ph(t-1))$$

$$i(t) = \sigma(Vx(t) + Qh(t-1))$$

$$c(t) = c(t-1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t-1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors  $x(t)$ , hidden activation vectors  $h(t)$ , memory cell state vectors  $c(t)$ , gate activation vectors  $z(t), i(t), o(t)$ , weight matrices  $P, Q, R, U, V, W$ . Let  $L(t) = L(y(t), \hat{y}(t))$  denote the loss at time  $t$  and let  $L = \sum_{t=1}^T L(t)$  denote the total loss. We use denominator-layout convention, i.e.,  $\frac{\partial L}{\partial c(t)}$  is a column vector. The diag operator turns a vector into a diagonal matrix, i.e.,  $\text{diag}((1, 1)^\top) = I \in \mathbb{R}^{2 \times 2}$  and  $\odot$  denotes Hadamard's product. Which of the following statements are true?

- ☐ a. The fact that  $\frac{\partial h(t)}{\partial c(t)} = \text{diag}(o(t) \odot \tanh'(c(t)))$  is a diagonal matrix is the key to how LSTM solves the vanishing gradient problem.
- ☒ b. Changing the definition of the cell state to  $c(t) = c(t-1) \odot f(t) + z(t) \odot i(t)$ , where  $f(t)$  is another sigmoid-activated gate reintroduces the vanishing gradient problem to a certain extent. ✓
- ☐ c. The gradient of the loss with respect to the hiddens was originally proposed to be approximated by  $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t+1)}{\partial h(t)}$ .
- ☐ d. The gradient of the loss with respect to the memory cell is  $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t+1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$ .

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: Changing the definition of the cell state to

$c(t) = c(t-1) \odot f(t) + z(t) \odot i(t)$ , where  $f(t)$  is another sigmoid-activated gate reintroduces the vanishing gradient problem to a certain extent., The gradient of the loss with respect to the memory cell is  $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t+1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$ .



**Question 15**

Correct

Mark 1.00 out of 1.00

Which statements about LSTM are true?

- ☐ a. The forget gate should be initialized with negative bias values.
- ☒ b. "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network. ✓
- ☒ c. The focused LSTM network feeds the input vector  $x(t)$  only into the block input  $z(t)$  and not into input gate  $i(t)$  or output gate  $o(t)$ . ✓
- ☐ d. The inputs to the lightweight LSTM have to fulfill the Markov property.
- ☒ e. Most LSTM networks used in practice feature a forget gate. ✓
- ☐ f. To counter the drift effect, the input gate should be initialized with positive values.
- ☐ g. The forget gate solves the vanishing gradient problem.
- ☒ h. An LSTM with coupled input gate and forget gate, i.e.,  
 $c(t) = (1 - i(t)) \odot c(t - 1) + i(t) \odot z(t)$ , leads (alongside other changes) to the gated recurrent unit (GRU). ✓

Your answer is correct.

The correct answers are: Most LSTM networks used in practice feature a forget gate., An LSTM with coupled input gate and forget gate, i.e.,  $c(t) = (1 - i(t)) \odot c(t - 1) + i(t) \odot z(t)$ , leads (alongside other changes) to the gated recurrent unit (GRU)., The focused LSTM network feeds the input vector  $x(t)$  only into the block input  $z(t)$  and not into input gate  $i(t)$  or output gate  $o(t)$ ., "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network.

**Question 16**

Correct

Mark 1.00 out of 1.00

Which statements about LSTM dropout are correct?

- ☐ a. LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients.
- ☒ b. Dropout is a regularization technique that randomly sets activations to zero during the forward pass. ✓
- ☐ c. It is especially important to apply dropout when evaluating the model.
- ☒ d. The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content. ✓
- ☒ e. LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability. ✓

Your answer is correct.

The correct answers are: LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability., The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content., Dropout is a regularization technique that randomly sets activations to zero during the forward pass.

**Question 17**

Correct

Mark 1.00 out of 1.00

Which of the following statements are true regarding the Mamba architecture?

- ☐ a. Mamba's selective SSM differs from traditional SSMs because its (**A**, **B**, **C**) functions are independent of the input.
- ☒ b. Mamba uses pre-activation forget (sd), input (sb), and output (sc) gates - these gates control the flow of information within the Mamba block. ✓
- ☒ c. Mamba is considered a selective State Space Model (SSM) ✓
- ☐ d. Mamba's memory requirements scale quadratically  $\mathcal{O}(n^2)$  with input sequence length.
- ☒ e. The Mamba architecture combines the H3 block of SSM with the MLP block structure of transformer. ✓

Die Antwort ist richtig.

The correct answers are: Mamba is considered a selective State Space Model (SSM), The Mamba architecture combines the H3 block of SSM with the MLP block structure of transformer., Mamba uses pre-activation forget (sd), input (sb), and output (sc) gates - these gates control the flow of information within the Mamba block.